



www.mrao.cam.ac.uk/~wh260/dark_energy_POS.pdf

Dark Energy

The mysterious “driving force” of the Universe

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First things first...

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Dark Energy $\neq$ Dark Matter

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- ▶ Dark matter: 1930's
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Universal consequences of gravity

Thought experiment

Universal consequences of gravity

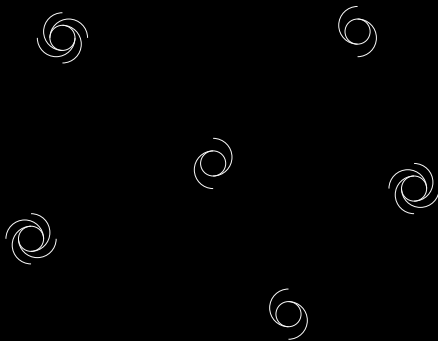
Thought experiment



Universal consequences of gravity

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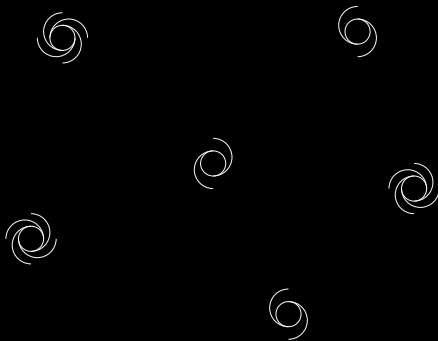
- Place galaxies at rest.



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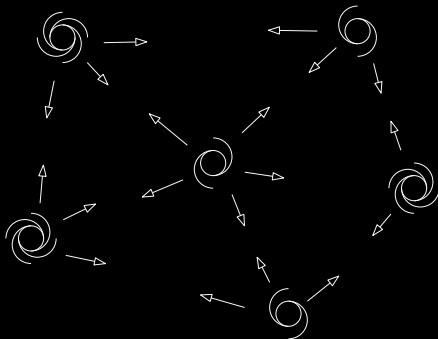
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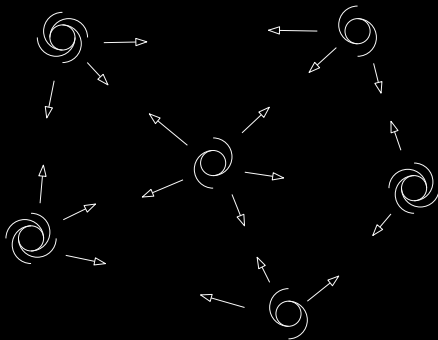
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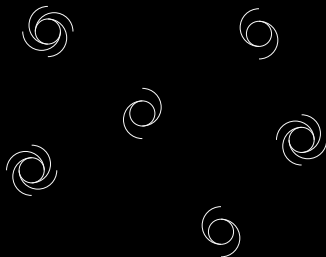
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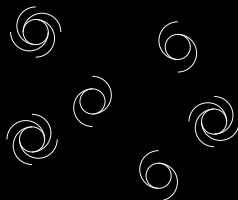
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Universal consequences of gravity

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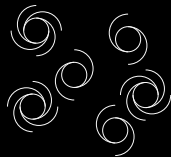
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The consequences for the universe

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- ▶ This suggests two possibilities...

The evolution of the universe

Two possibilities

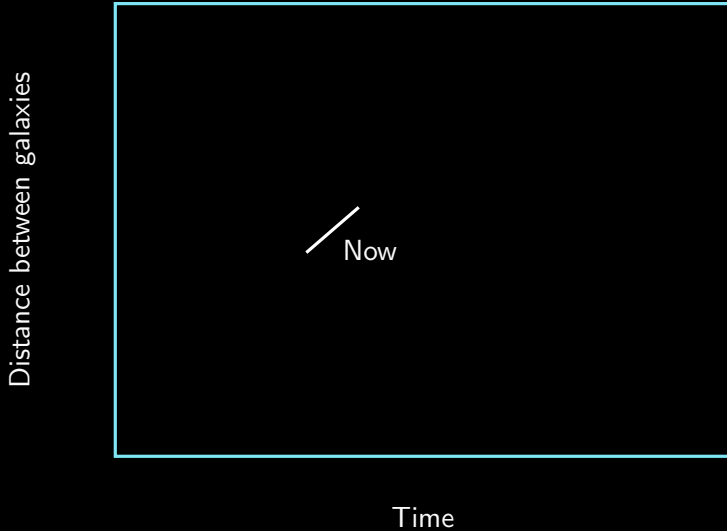
Distance between galaxies



Time

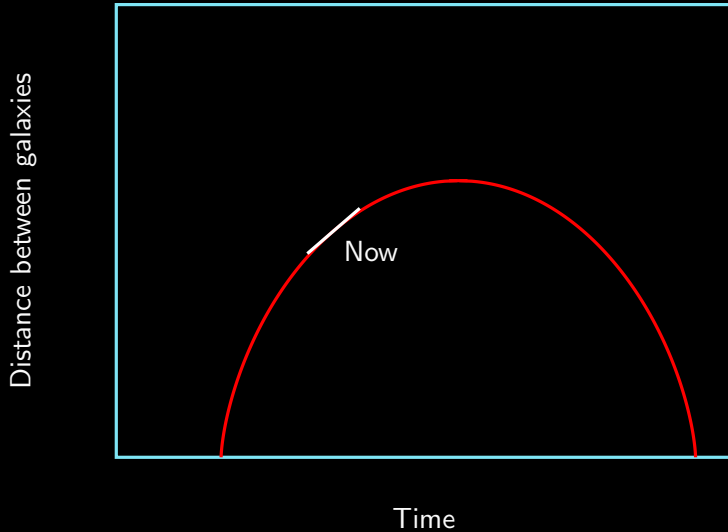
The evolution of the universe

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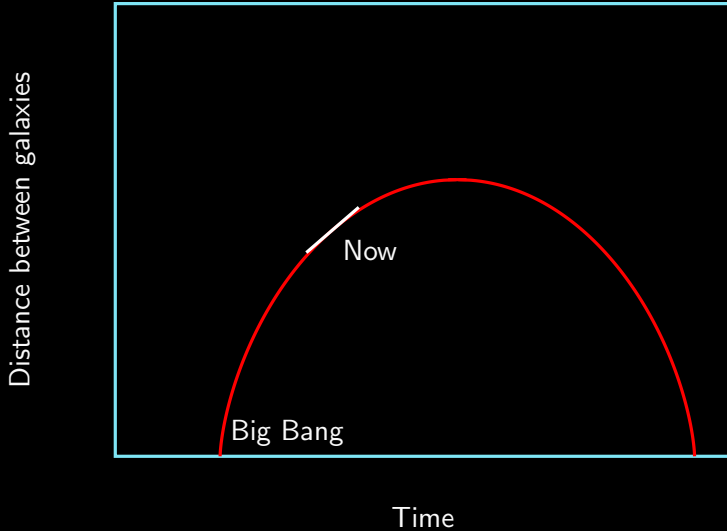
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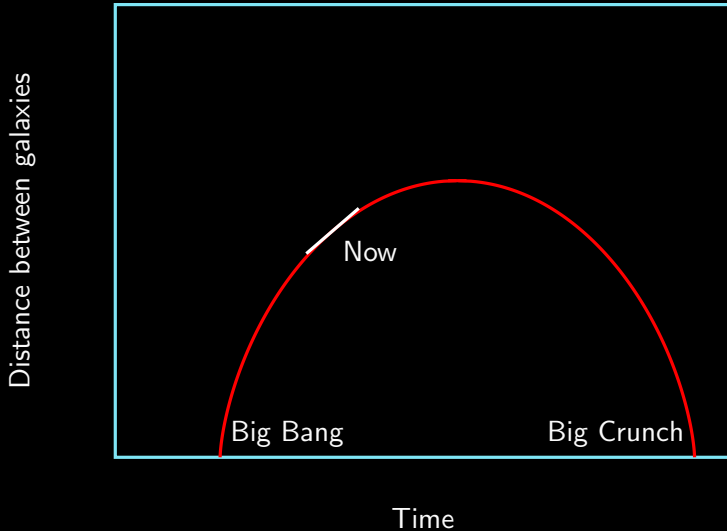
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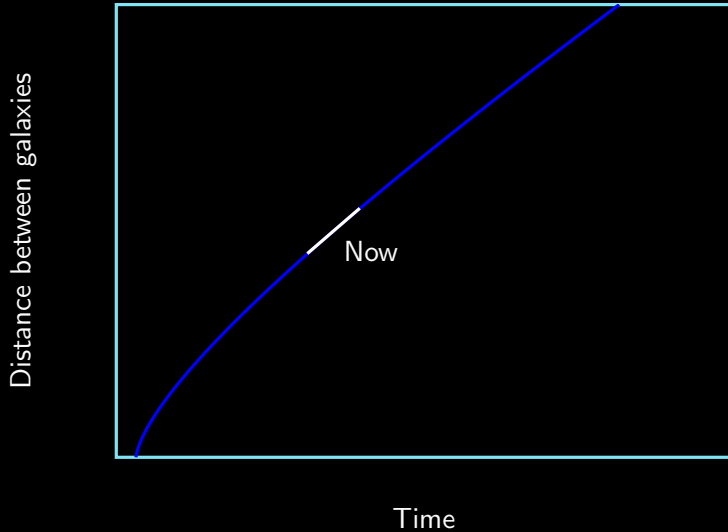
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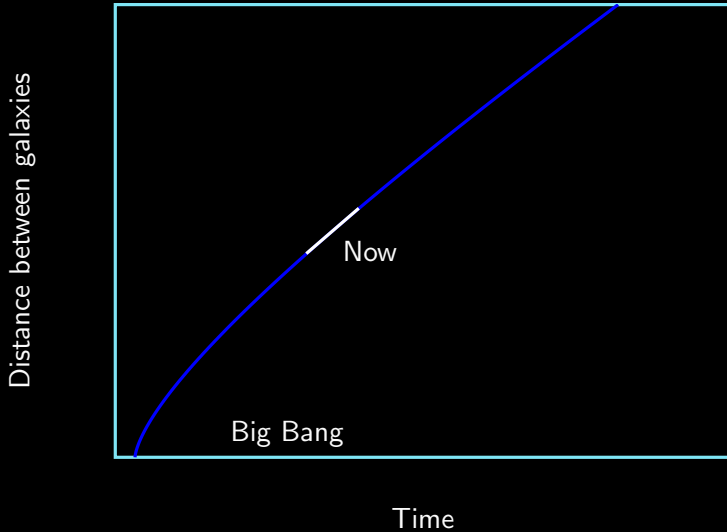
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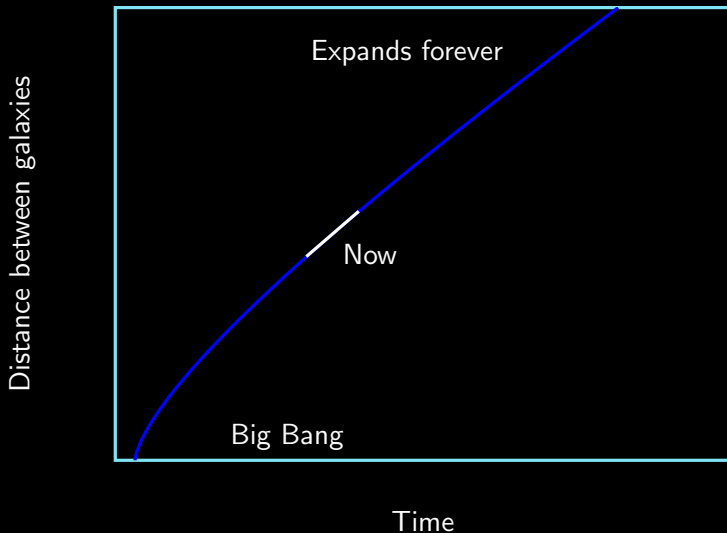
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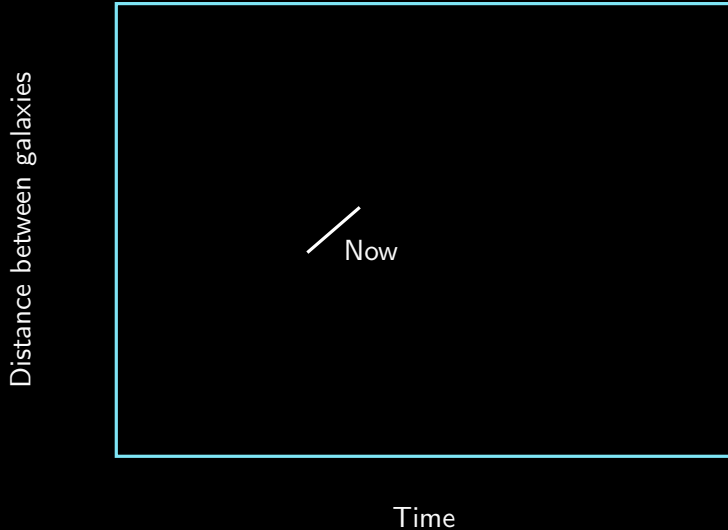
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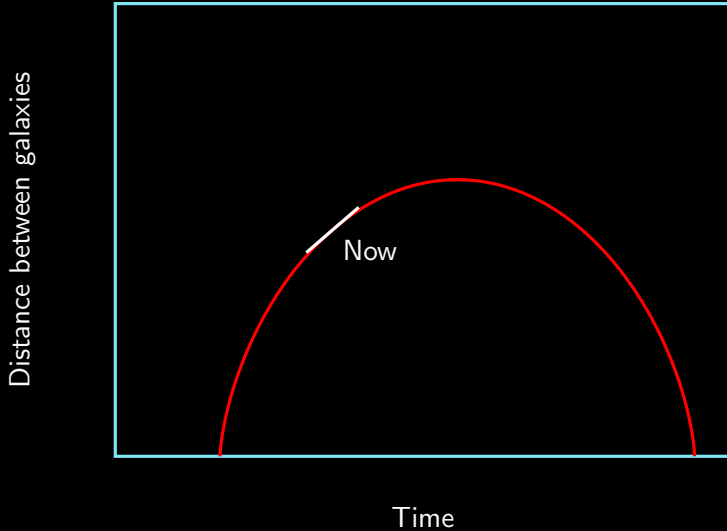
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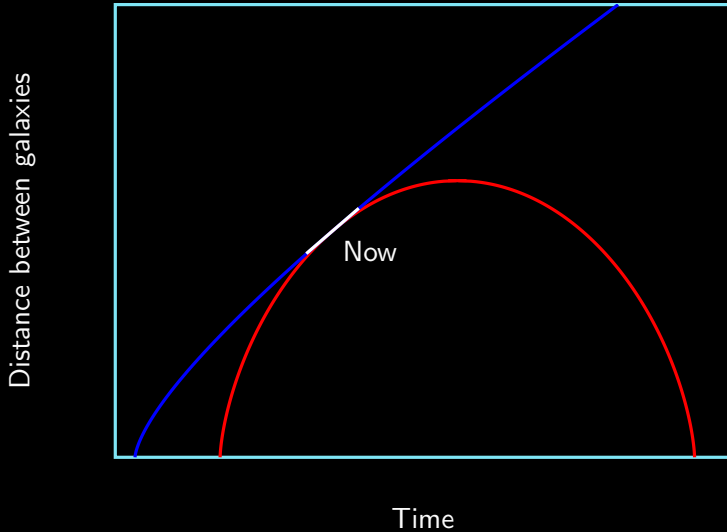
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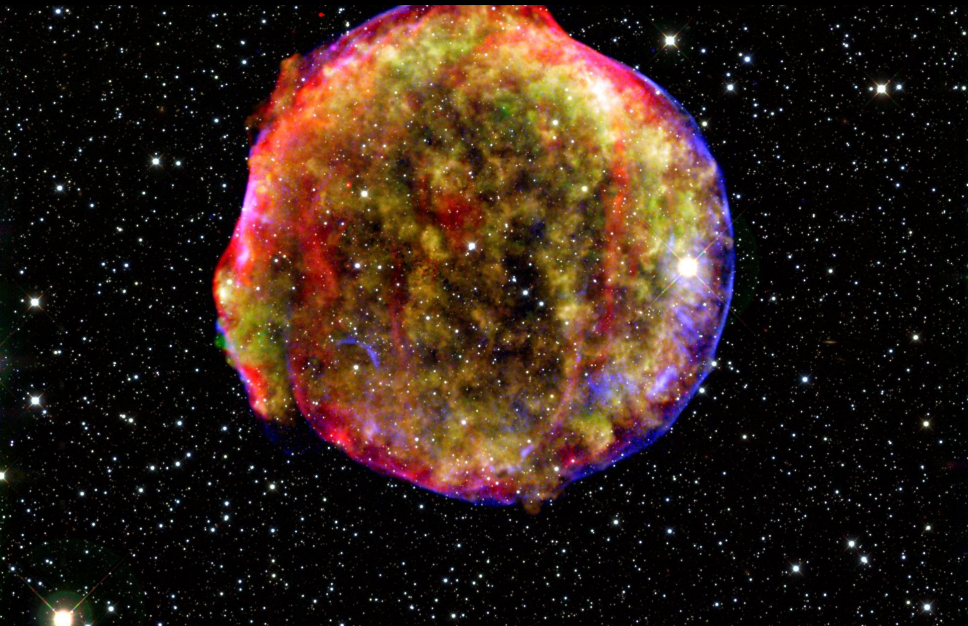
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Two possibilities



Exploding stars

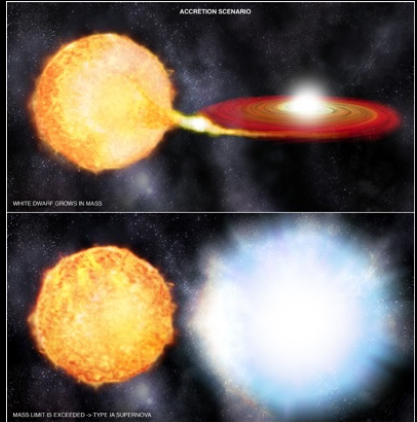
Type IA Supernovae



Type IA Supernovae

Standard Candles

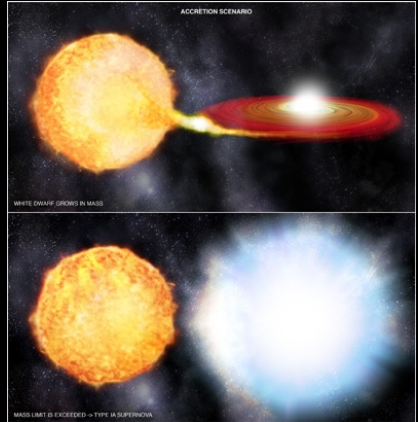
- Produces supernova with specific signature.



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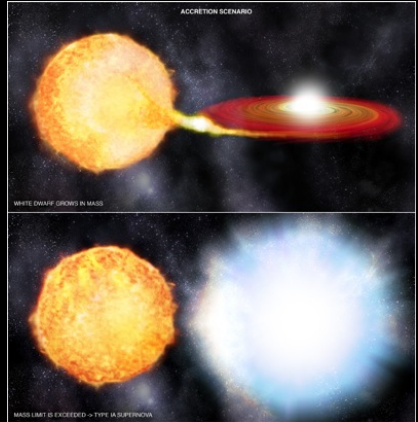
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- ▶ Brightness $\Rightarrow$ distance.



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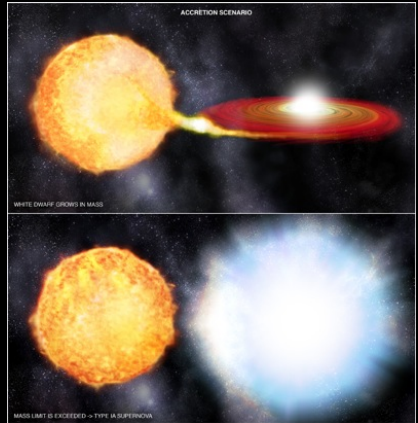
- ▶ Produces supernova with specific signature.
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- ▶ Redshift $\Rightarrow$ epoch (time).



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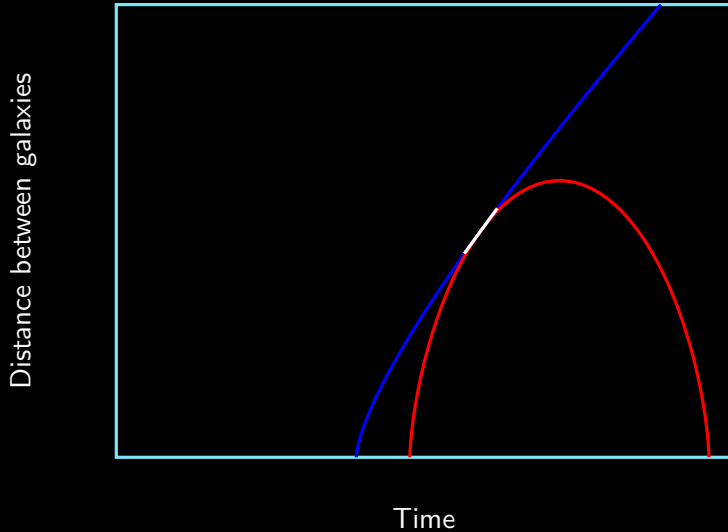
Standard Candles

- ▶ Produces supernova with specific signature.
- ▶ Brightness $\Rightarrow$ distance.
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- ▶ Accretion in binary star system.



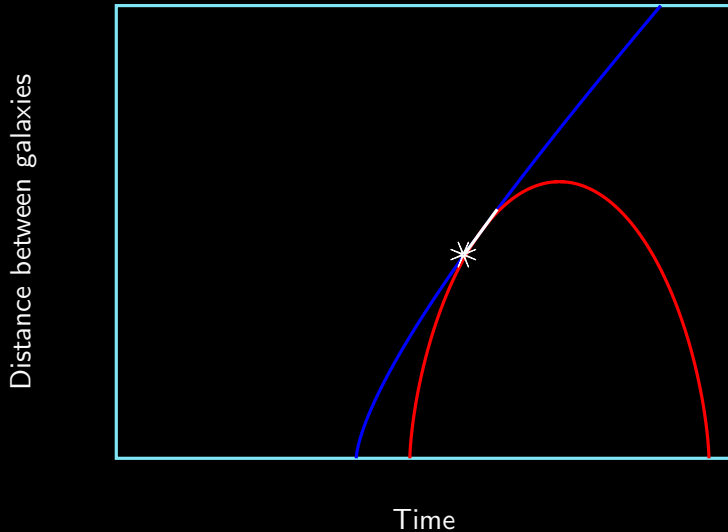
The evolution of the universe

Which is it?



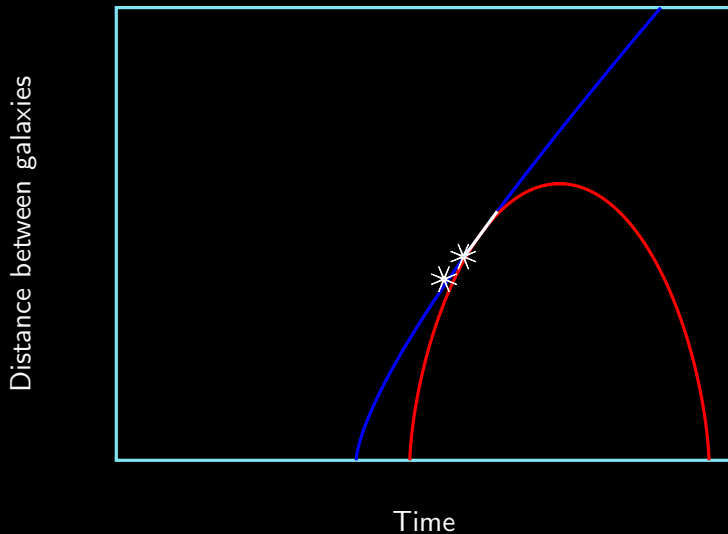
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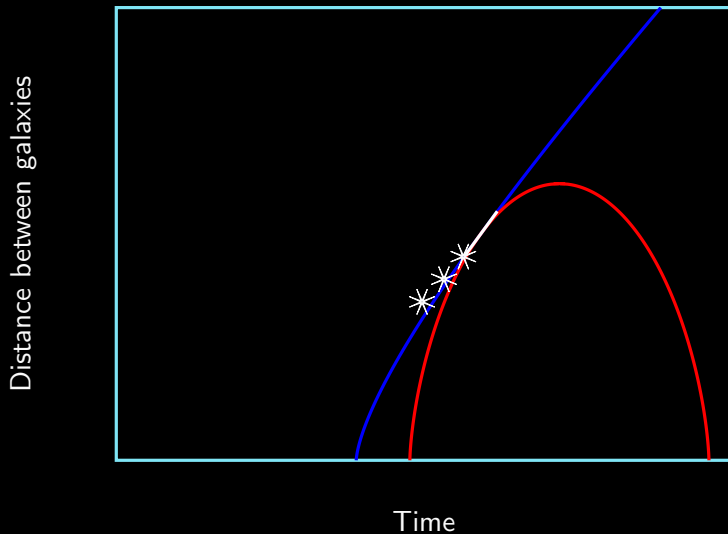
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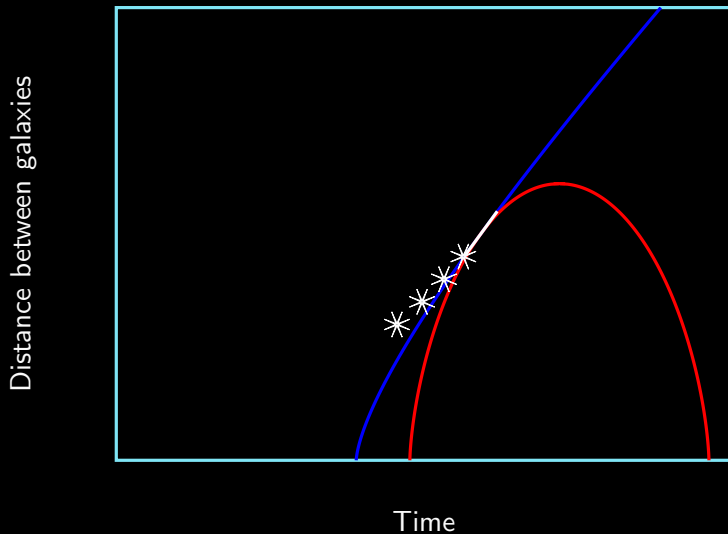
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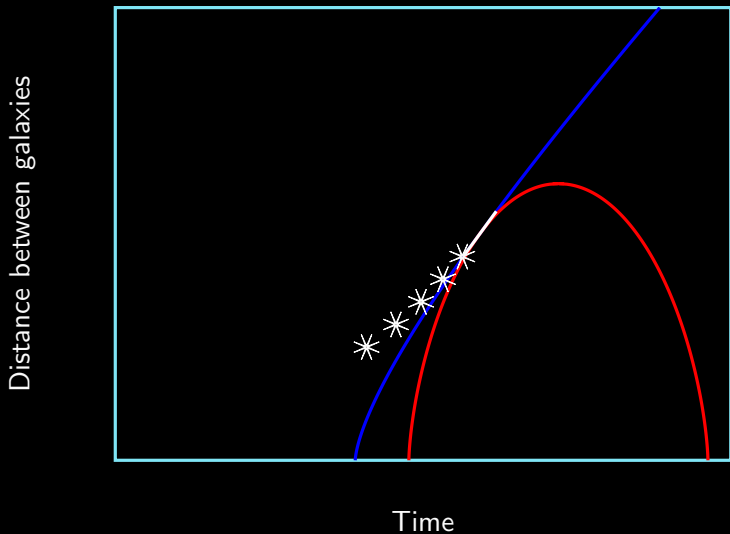
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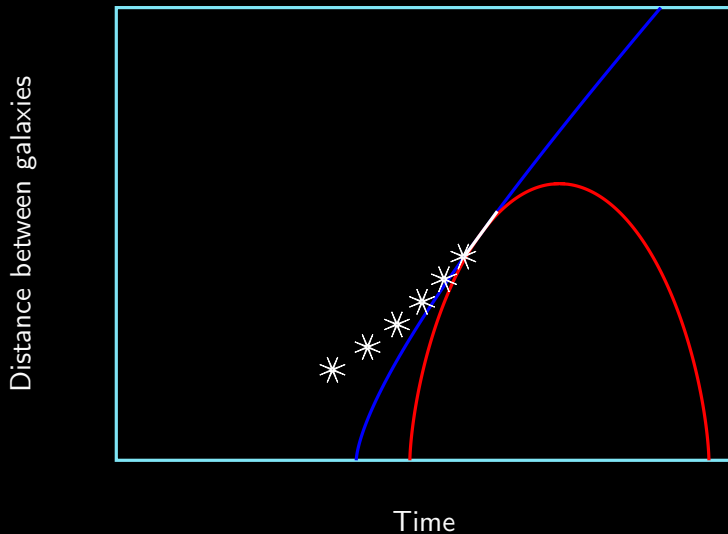
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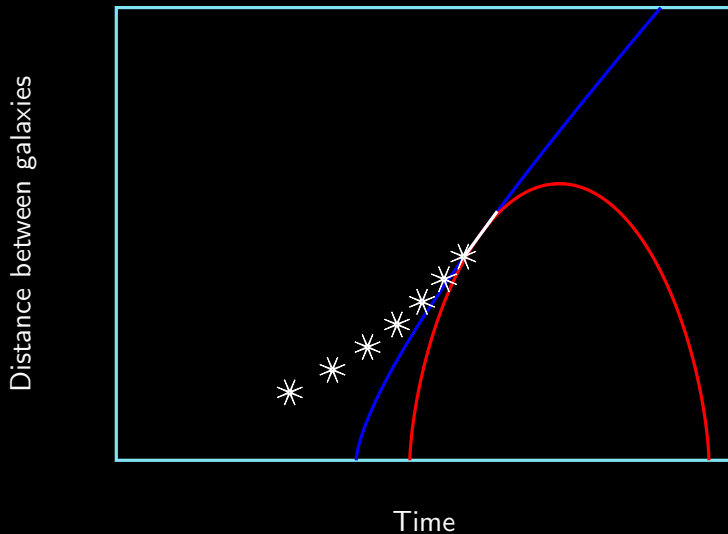
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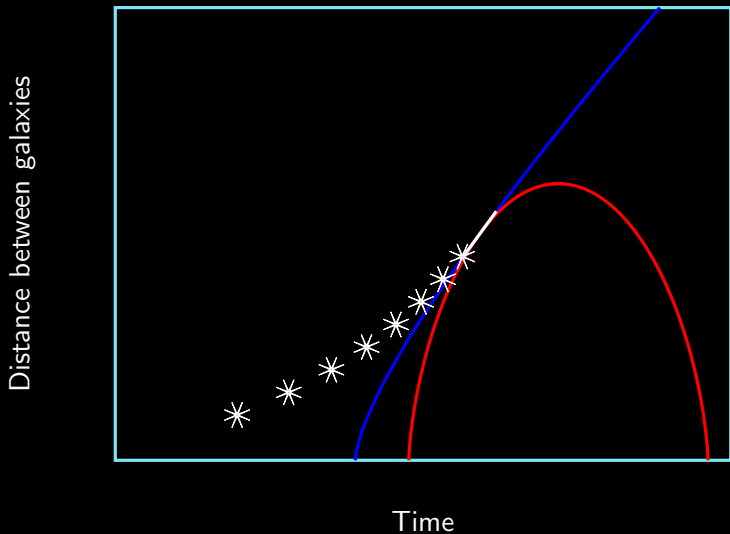
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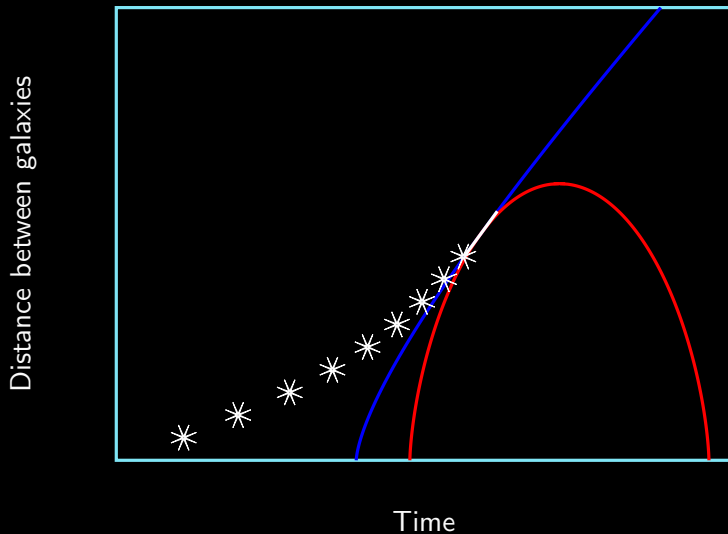
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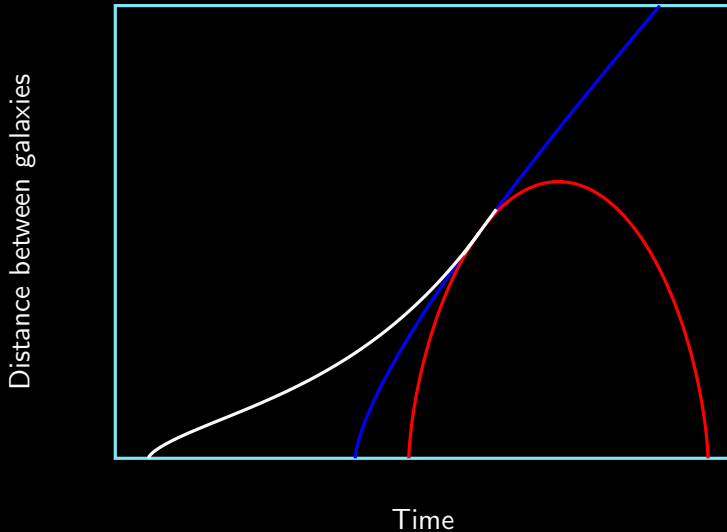
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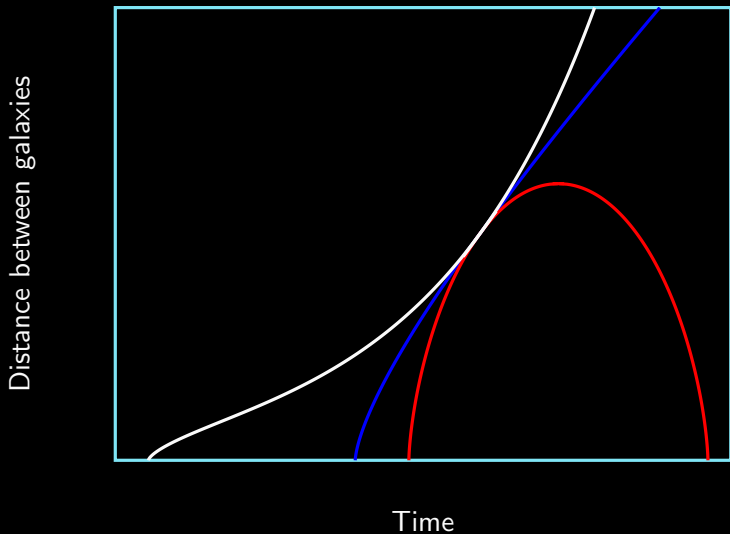
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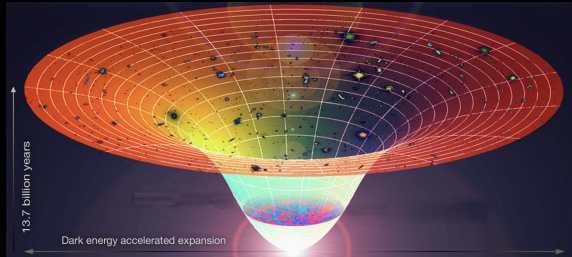
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Which is it?



Accelerated expansion

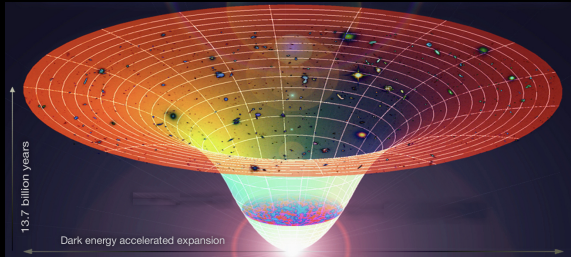
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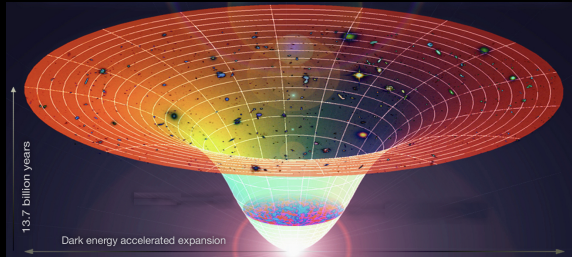
- ▶ At some point in the past (≈ 4 bya) the universe's expansion started speeding up!



Accelerated expansion

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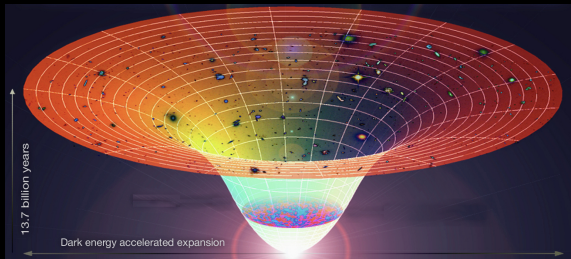
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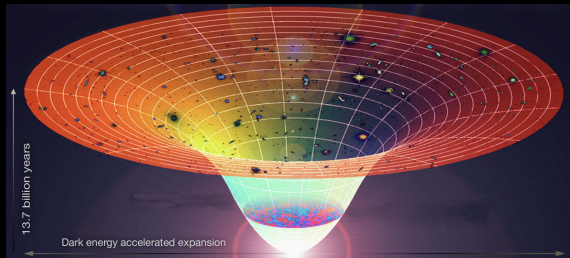
- ▶ At some point in the past (≈ 4 bya) the universe's expansion started speeding up!
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- ▶ As if some “force” is driving the increased expansion.



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- ▶ At some point in the past (≈ 4 bya) the universe's expansion started speeding up!
- ▶ In stark contrast to what we predicted.
- ▶ As if some “force” is driving the increased expansion.
- ▶ Or, as if space itself has some inherent “springyness” pushing everything apart.



Dark Energy

- Dark energy is a form of energy that is hypothesized to be responsible for the observed acceleration of the expansion of the universe.
- It is a mysterious force that counteracts the attractive force of gravity, causing the universe to expand at an increasing rate.
- Dark energy is estimated to make up about 68% of the total energy density of the universe.
- The existence of dark energy was first proposed in 1998 by two teams of astronomers who were studying the expansion of the universe using distant supernovae.
- Since then, numerous other observations, including the cosmic microwave background and large-scale structure surveys, have provided strong evidence for the existence of dark energy.
- Despite its name, dark energy is not "dark" in the sense of being invisible or undetectable. It is simply a name given to a form of energy that we do not yet fully understand.
- One of the leading theories for the nature of dark energy is the cosmological constant, a term introduced by Albert Einstein in his equations of general relativity.
- Another leading theory is the quintessence model, which proposes that dark energy is a dynamic field that varies in space and time.
- The study of dark energy is one of the most active areas of research in cosmology, and it is expected that future observations will provide a deeper understanding of this mysterious force.

Dark Energy

- ▶ We attribute this accelerated expansion to a quantity we call *Dark Energy*.

Dark Energy as a prediction

In general, if you encounter a discrepancy between theory and observation, there are two options:

Dark Energy as a prediction

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Dark Energy as a prediction

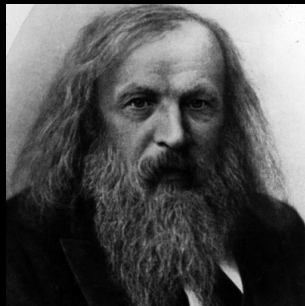
In general, if you encounter a discrepancy between theory and observation, there are two options:

- ▶ Modify the theory
- ▶ Predict a new object

Example 1.

The Periodic Table

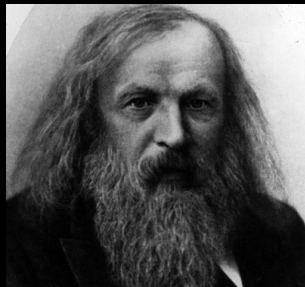
- ▶ In 1865, Mendeleev proposed that elements be ordered into the periodic table.



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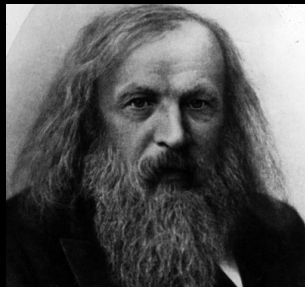


	Gruppe I. — H ⁰	Gruppe II. — H ⁰	Gruppe III. — H ⁰	Gruppe IV. H ¹ H ⁰	Gruppe V. H ¹ H ⁰	Gruppe VI. H ¹ H ⁰	Gruppe VII. H ¹ H ⁰	Gruppe VIII. — H ⁰
1	H=1							
2	Li=7	Be=9,4	B=11	C=12	N=14	O=16	F=19	
3	Na=23	Mg=24	Al=27,5	Si=28	P=31	S=32	Cl=35,5	
4	K=39	Ca=40	—=44	Ti=48	V=51	Cr=52	Mn=55	Fe=56, Co=59, Ni=59, Cu=63.
5	(Ca=63)	Zn=65	—=68	—=72	As=75	Se=78	Br=80	
6	Rb=85	Str=87	Yt=88	Zr=90	Nb=94	Mo=96	—=100	Ra=104, Rn=104, Po=106, Ag=108.
7	(Ag=108)	Cd=112	In=115	Sn=118	Sb=122	Te=125	J=127	
8	Cs=133	Ba=137	Pd=138	Ce=140	—	—	—	
9	(—)	—	—	—	—	—	—	
10	—	—	—	—	—	—	—	
11	(Au=197)	Hg=200	Tl=204	Pb=207	Bi=208	—	—	
12	—	—	—	Th=231	—	U=240	—	

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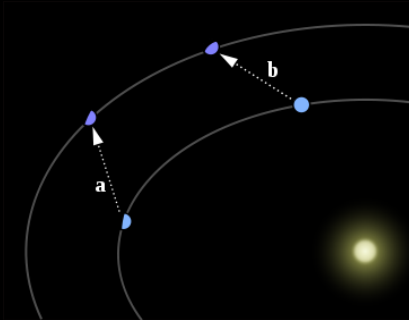
- ▶ In 1865, Mendeleev proposed that elements be ordered into the periodic table.
- ▶ He left gaps in the table, where his theory predicted there should be new matter.
- ▶ When these elements were discovered, he was hailed as a scientific hero.



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7	(Ag=108)	Cd=112	In=115	Su=118	Sh=122	Tu=125	J=127	
8	Ce=123	Ba=127	YDi=128	PCe=140	—	—	—	
9	(—)	—	—	—	—	—	—	
10	—	—	YEr=178	YLa=180	Ta=182	W=184	—	Os=195, Ir=197, Pt=198, Au=199.
11	(Au=199)	Hg=200	Ti=204	Pb=207	Bism=208	—	—	
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Example 2.

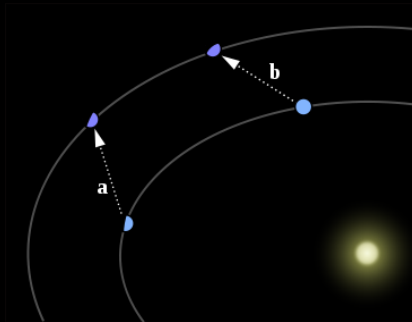
Orbits of Uranus and Mercury



- In 1841, it was noticed that Uranus' orbit was slightly off from predictions.

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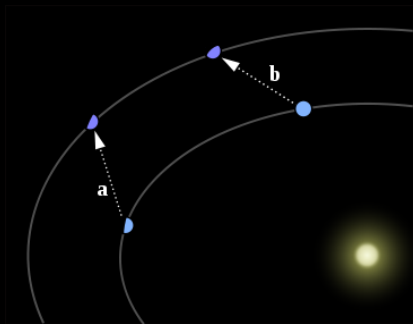
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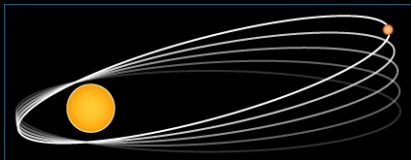
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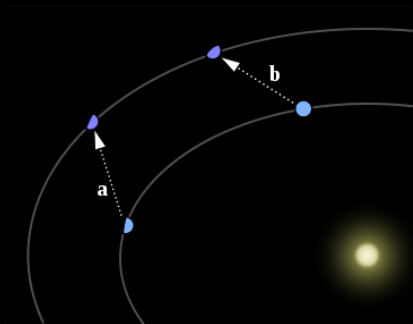
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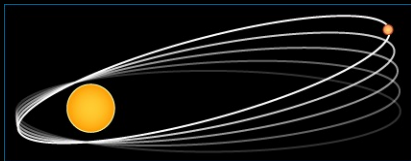
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- ▶ In 1859, a similar problem emerged with the orbit of mercury
- ▶ This issue wasn't solved until Einstein's theory of general relativity superseded Newton's gravity in 1915.

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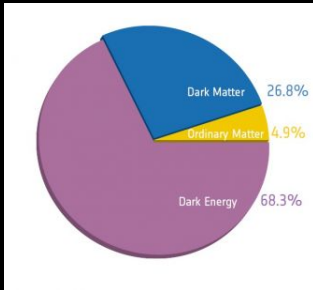
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Anything producing a large scale, diffuse, (negative) pressure.

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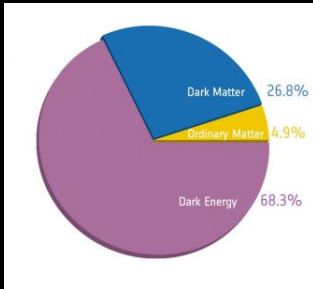
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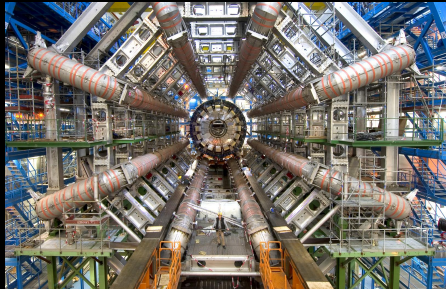
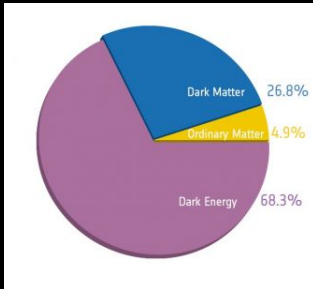
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- ▶ Nope. Factor of 10^{120} too much energy in Higgs field.

What could this all pervading energy be?

Common interpretation

- [illegible]

Dark Energy...

...or Cosmological Constant?

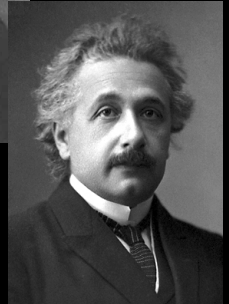
- ▶ Einstein's gravity can be repulsive at large scales.



Dark Energy...

...or Cosmological Constant?

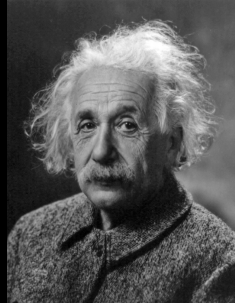
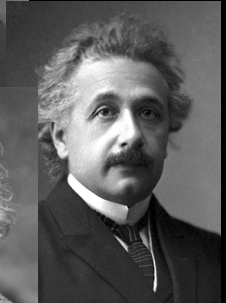
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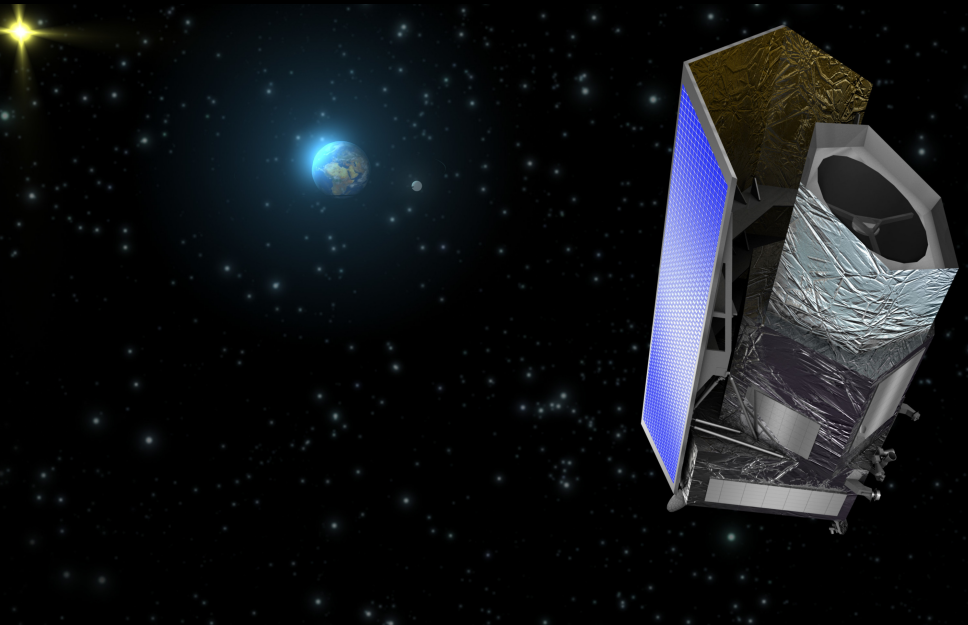
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- ▶ Hubble Bubble: “Optical illusion”
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- ▶ Many more...

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- ▶ Hubble Bubble: “Optical illusion”
- ▶ More drastic alterations of gravity.
- ▶ Many more. . .
- ▶ What is needed is more *data*.

The Euclid Satellite

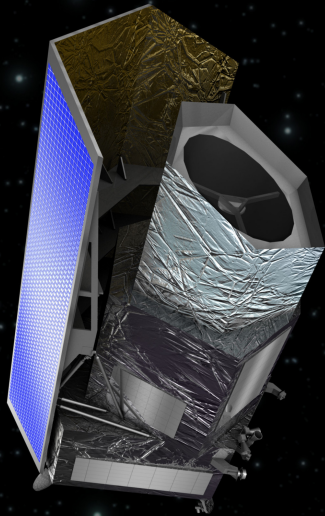
Energy vs. Cosmological Constant



The Euclid Satellite

Energy vs. Cosmological Constant

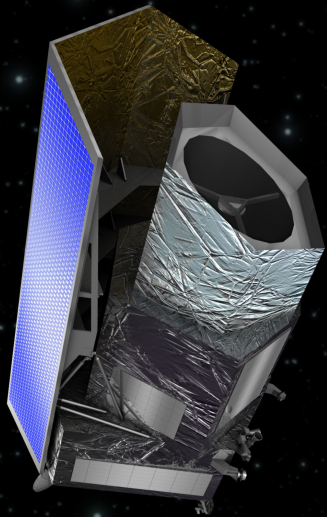
► Launch date $\approx$ 2020



The Euclid Satellite

Energy vs. Cosmological Constant

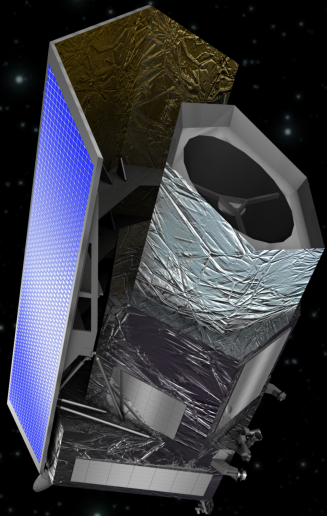
- ▶ Launch date ≈ 2020
- ▶ Build up an accurate 4D picture of the visible universe



The Euclid Satellite

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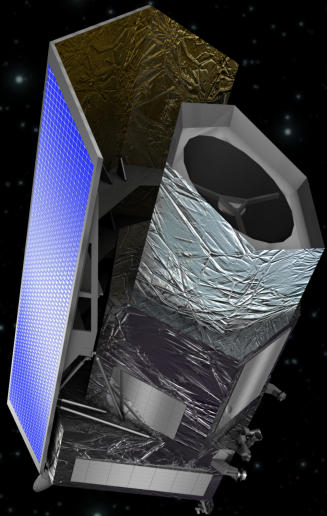
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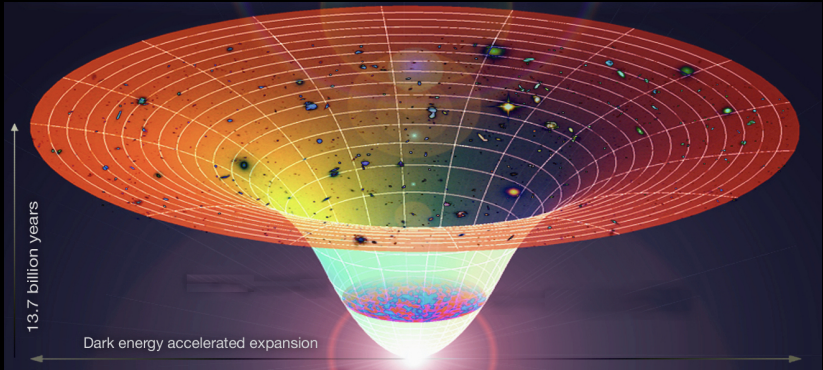
The Euclid Satellite

Energy vs. Cosmological Constant

- ▶ Launch date ≈ 2020
- ▶ Build up an accurate 4D picture of the visible universe
- ▶ By measuring cosmic shear around galaxies and clusters, will build up a map of matter
- ▶ Will see any variation in dark energy across space and time, enabling us to build a better understanding of it.

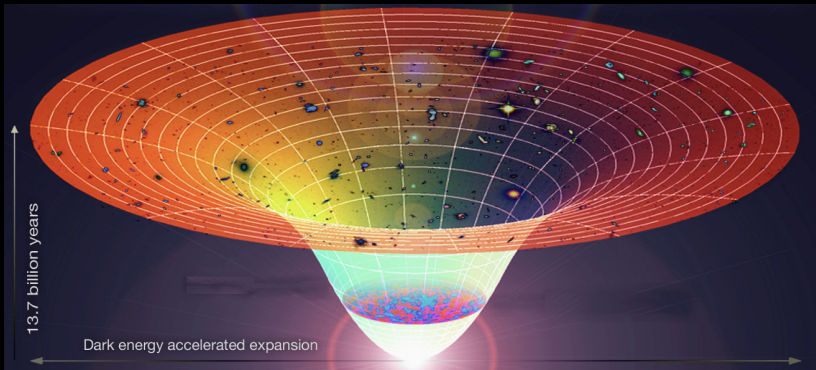


Summary



- Dark energy is the force behind the accelerating universe.

Summary



- ▶ Dark energy is the force behind the accelerating universe.
- ▶ This is an exciting “hint” of undiscovered physics.

Further Reading

Hubble & Type IA http://hubblesite.org/hubble_discoveries/dark_energy/

Euclid Satellite <http://sci.esa.int/euclid/42267-science/>

BBC Radio <http://www.bbc.co.uk/programmes/p003k9g5>